

SD-8400 Solar Inverter

Cooling System, Operations & Fault Response Manual

DOCUMENT	SD-OM-8400
VERSION	3.9
EFFECTIVE	April 2, 2026
STATUS	Approved — Synthetic
MANUFACTURE R	Solara Dynamics (Synthetic)

8. Cooling-Fan Monitoring

The SD-8400 compares each commanded fan-speed signal with the corresponding feedback signal and records a derived variance for equipment-health diagnostics.

8.1 Interpreting command and feedback

FAN COMMAND	Controller-requested fan speed; it does not prove mechanical rotation.
FAN FEEDBACK	Reported speed; it may be affected by fan, connector, wiring, or sensing conditions.
DERIVED VARIANCE	Absolute command/feedback difference; it identifies disagreement, not the failed component.

8.2 Authorized response boundary

- Use site-approved procedures before inspecting or servicing cooling equipment.
- Do not bypass fan supervision, force a fan command, or clear a maintenance hold solely to restore output.
- Opening an enclosure or contacting internal components requires the applicable isolation and lockout/tagout process.

Safety note: This synthetic manual does not replace site electrical-safety, isolation, or lockout/tagout requirements.

8.4 Fault A12 — Cooling Fan Feedback Variance

Fault A12 is asserted when the derived cooling-fan command/feedback variance exceeds the configured 12-percentage-point threshold during the controller diagnostic window. The inverter enters Maintenance Hold so the cooling system can be inspected before return to service.

Expected behavior: A12 establishes that commanded and reported fan speed disagreed beyond the configured limit. It does not, by itself, prove a failed fan. Obstruction, connector or wiring conditions, feedback sensing, and intermittent mechanical drag remain possible until inspected.

Required checks

- 1 Keep the inverter in Maintenance Hold and confirm the event and diagnostic window with the control room.
- 2 Inspect accessible intake screens, exhaust openings, fan guards, and the exterior cooling-air path for obstruction, contamination, or damage.
- 3 Compare fan command and fan feedback. Treat missing, unstable, or persistently divergent feedback as unresolved.
- 4 Use the approved site procedure before any internal inspection. Do not open energized compartments for this check.
- 5 Apply the approved return-to-service checklist after corrective work; do not clear A12 solely because the alarm is acknowledged.

Evidence interpretation: High ambient temperature can increase fan demand, but ambient temperature alone does not explain excessive command/feedback variance or establish component failure.